

BUS63-8

Financial Instruments

Albert School — 22 hours

Preface

In **Valuation & Investment Analysis** (BUS43-3) you learned to price a **firm**. You built a DCF from its statements, triangulated it against multiples, stressed the two assumptions that move the answer, and wrote a buy / hold / sell thesis you could defend before a practitioner. The output of that course was a **view on one company**.

This course changes the unit of analysis. A view on a company has to be **expressed** through something you can actually buy, hold, and sell — a share, a bond, an option, a fund — and each of those instruments has its own price, its own risks, and its own mathematics. “I think this firm is undervalued” becomes “buy the equity”; “I expect rates to fall” becomes “lengthen duration”; “I want the upside but not the downside” becomes “buy a call”; “I am exposed to the dollar” becomes “sell a currency forward.” BUS63-8 is the **analytical toolkit** for that translation: how each instrument family is priced from data, how its risk is measured, and how instruments combine into a portfolio aligned with a goal.

It is the analytical half of the S6 Financial Markets capstone pair. Its sibling, **Investment Philosophies** (BUS63-9), asks **when and why** to deploy each approach — value versus growth, active versus passive, the behavioural traps of real investors. This course asks the prior question: **how does each instrument actually work, and what is it worth?** You cannot sensibly choose a philosophy over tools you cannot operate.

Three habits carry through every chapter. First, **everything is priced as present value** — a bond, an option, a portfolio’s expected return are all the same discounting logic you already own, applied to different cash flows. Second, **return is never read without risk beside it** — every instrument and every portfolio in this book is judged on the two together, never the headline return alone. Third, **the data is real and so are its limits** — you will compute YTM’s, durations, Sharpe ratios, and optimal weights from actual market data in Python, and the most important thing you will learn about the most elegant model (mean-variance optimisation) is precisely where it breaks. By the end, a single reflex should fire at any instrument: **what are its cash flows, what moves its price, and what does adding it do to the portfolio I already hold?**

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1 The four instrument families

1.1 One firm, four ways to own a claim on it

Take a single company — say a large, profitable manufacturer. You can own a **share** of it and receive whatever is left after everyone else is paid: unlimited upside, no promised payment, last in line if it fails. You can lend it money by buying its **bond** and receive fixed coupons and your principal back at maturity: capped upside, a contractual promise, ahead of the shareholders if it fails. You can buy an **option** on its shares and obtain the right — but not the obligation — to buy at a fixed price, paying a small premium for leveraged exposure to its price moves. Or you can buy a **fund** that holds this firm alongside hundreds of others, converting a single bet into diversified exposure. Same underlying company; four completely different risk-return profiles, because each instrument occupies a different place in the firm's capital structure and a different point on the risk-return map.

1.2 The families and their profiles

Definition 1 (the four instrument families) Financial instruments fall into four families, distinguished by the **kind of claim** they represent:

- **Equities** — ownership claims on a firm's **residual** value. The holder receives dividends and price appreciation, has no maturity and no promised payment, and ranks **last** in liquidation. Highest expected return, highest risk.
- **Debt** — contractual claims to **fixed or floating** payments and principal at maturity. Senior to equity, exposed to **interest-rate** and **credit** risk. Moderate, more predictable return.

- **Derivatives** — contracts whose value **derives from an underlying** asset (futures, options, swaps). Used to **hedge** an existing risk or to take **leveraged** exposure; can have very large gains or losses relative to the capital committed.
- **Funds** — **pooled vehicles** (mutual funds, ETFs) holding a portfolio of the above. Deliver diversification and either **active** or **passive** management, for a fee.

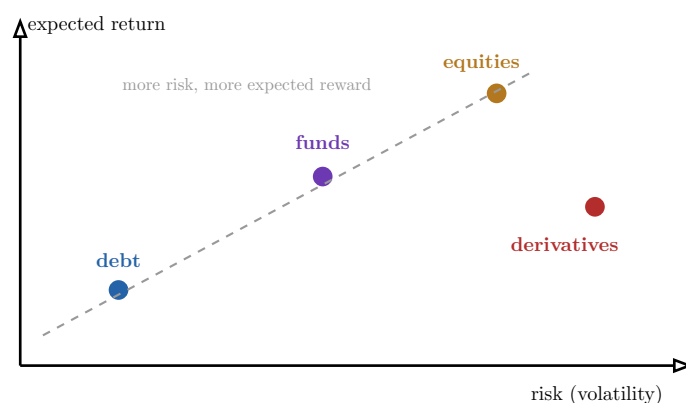


Figure 1: The risk-return map of the four families. Debt sits low-risk / low-return; equities high / high; diversified funds in between. Derivatives sit **off** the line: their position depends entirely on how they are used — a hedge **reduces** risk, while a leveraged speculative position pushes far to the right. The map, not the label, is what matters when matching an instrument to a use case.

Remark Derivatives are the family students mis-file most often. A derivative is not **inherently** high-risk: the **same** call option is a prudent hedge for a producer locking in a sale price and a reckless gamble for a speculator with no underlying exposure. Classify a derivative by **how it sits against an existing exposure**, not by the contract type alone.

Worked example — Matching the family to the investor. A pension fund needing predictable cash to pay retirees buys **investment-grade bonds** — income and capital preservation. A venture investor seeking multiples of capital buys **equity** — residual upside. An airline fearing a fuel-price spike buys **futures** on jet fuel — a hedge, not a bet. A retail saver wanting cheap diversified exposure buys an **index ETF** — a fund. Each chose the family whose risk-return profile matches the **job to be done**, not the one with the highest headline return.

Exercises

1. Classify a given set of instruments into the four families, and for each state its position in the capital structure, its risk-return profile, and a typical investor use case.
2. For one derivative, describe a use that makes it a **hedge** and a use that makes it a **speculation**, and explain why the contract is identical but the risk profile is not.

2 Equity analysis via a quantified fundamental score

2.1 From a paragraph of opinion to a number you can rank

In BUS43-3 you wrote a thesis on **one** firm in prose. But an asset manager screening five hundred stocks cannot write five hundred essays. The discipline here is to compress the fundamental case

into a **single comparable number** — a score — computed identically for every name, so the whole universe can be **ranked** and the analyst's scarce attention spent on the top and bottom of the list. The art is choosing inputs that actually predict quality and combining them honestly.

2.2 The three input groups

Definition 2 (quantified fundamental score) A **quantified fundamental score** rates an equity by combining, from real market data, three groups of inputs into one comparable figure:

- **financial ratios** — profitability, leverage, and liquidity from the statements (e.g. ROE, net margin, debt/equity, current ratio): **is the business good?**
- **valuation multiples** — P/E, P/B, EV/EBITDA placing price against fundamentals (BUS43-3): **is it cheap relative to peers?**
- **market signals** — price- and volume-based indicators (momentum, relative strength): **what is the market already saying?**

Each raw input is **standardised** (e.g. into a percentile or z-score across the universe) so unlike units can be combined, then aggregated — typically a weighted average — into the final score.

Worked example — Scoring two stocks. Stock A: ROE 22%, net margin 15%, P/E 12, twelve-month momentum +18%. Stock B: ROE 6%, net margin 3%, P/E 30, momentum -10%. Standardised across the universe, A scores high on **quality** (top-quartile ROE and margin), high on **value** (below-median P/E), and high on **momentum**; B scores low on all three. A's composite ranks near the top, B near the bottom — and the manager looks hard at A first. The score did not **decide**; it **ordered** the search.

Common pitfall A composite score hides its own construction. Two traps recur. First, **correlated inputs double-count**: ROE, ROA, and net margin all measure profitability, so loading the score with three of them silently triples the weight on one idea. Second, **the weights are a hidden investment view**: putting 60% on value and 10% on momentum is a value strategy, whether or not you said so. Always state the weights, and check that the input groups measure genuinely different things before averaging them.

Exercises

3. Build a quantified fundamental score for a set of real equities from financial ratios, valuation multiples, and market signals; standardise each input, choose and justify the weights, and rank the names.
4. Show how the ranking changes when the weighting shifts from value-heavy to momentum-heavy, and explain what investment view each weighting encodes.

3 Bond pricing mechanics

3.1 A promise of fixed cash, and why its price moves

A bond is the simplest instrument in the book: a **stream of fixed cash flows** — coupons each period and the face value at maturity. If the cash is fixed, why does the price move at all? Because the **rate at which the market discounts that cash** moves. When prevailing interest

rates rise, a newly-issued bond pays more, so an existing bond paying the old, lower coupon is worth **less** — its price falls until its yield matches the market. The entire mechanics of bond pricing is this single inverse relationship between **price** and **yield**, made precise.

3.2 Price and yield to maturity

Definition 3 (bond price and yield to maturity) A bond's **price** is the present value of its cash flows — coupons C each period and face value F at maturity n — discounted at a single rate y , the **yield to maturity**:

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}.$$

The **YTM** is the one discount rate that makes this present value equal the bond's **observed market price**. It is the bond's internal rate of return if held to maturity, and it is found **numerically** — there is no closed form — by solving the equation above for y .

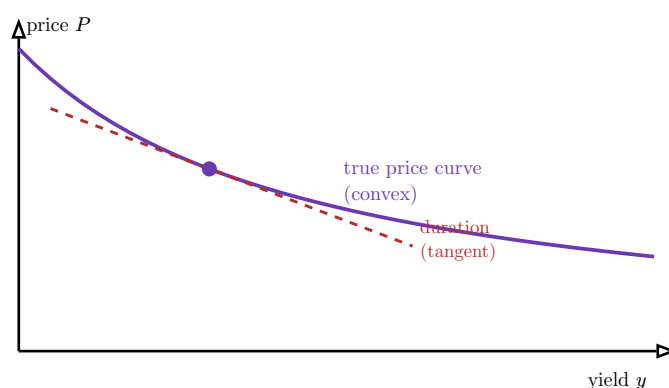


Figure 2: The price-yield relationship. Price **falls** as yield rises (the inverse relationship), and the curve is **convex** — bowed toward the origin. **Duration** is the slope of the tangent (dashed) at the current yield: a straight-line approximation that is accurate for small moves but, because the true curve bends away from it, **overstates** the price fall and **understates** the price rise for large moves. **Convexity** is the curvature that corrects this.

3.3 Duration: first-order interest-rate sensitivity

Definition 4 (duration) **Duration** measures how sensitive a bond's price is to its yield. **Macaulay duration** is the cash-flow-weighted average time to payment:

$$D = \frac{1}{P} \sum_{t=1}^n t \cdot \frac{C_t}{(1+y)^t}.$$

Modified duration $D^* = D/(1+y)$ converts this into a price sensitivity: for a small yield change Δy ,

$$\frac{\Delta P}{P} \approx -D^* \cdot \Delta y.$$

A modified duration of 7 means a +1% (100 bp) rise in yield costs roughly **7%** of the bond's value.

Worked example — Stress-testing a bond with duration. A bond trades at \$1,000 with modified duration 7. Rates rise 50 bp ($\Delta y = 0.005$). First-order estimate: $\Delta P/P \approx -7 \times 0.005 = -3.5\%$, a loss of about \$35. Longer-maturity, lower-coupon bonds have **higher** duration and so fall further for the same rate move — which is exactly why a portfolio manager expecting rate rises **shortens** duration, and one expecting cuts **lengthens** it.

3.4 Convexity: the second-order correction

Definition 5 (convexity) Convexity K is the **second-order** sensitivity of price to yield — the curvature of the price-yield curve. It refines the duration estimate:

$$\frac{\Delta P}{P} \approx -D^* \cdot \Delta y + \frac{1}{2}K \cdot (\Delta y)^2.$$

For standard (option-free) bonds $K > 0$, so the correction term is **positive** for a move in either direction: convexity **helps** the holder — prices rise more than duration predicts and fall less. It matters most for **large** yield moves, where the tangent diverges from the curve.

Common pitfall Duration is a local, linear approximation — do not trust it for large shocks. Because the price-yield curve is convex, duration alone **overstates** the loss from a big rate rise and **understates** the gain from a big fall. For the small moves of a normal day it is fine; for a stress scenario (± 200 bp) you must add the convexity term or, better, **fully reprice** the bond at the shocked yield. Reporting a stress-test loss from duration alone overstates the damage and is a classic novice error.

Exercises

5. For a real bond portfolio, compute each bond's YTM numerically, its Macaulay and modified duration, and its convexity, and aggregate to a portfolio duration and convexity.
6. Stress-test the portfolio under a range of yield shocks (e.g. ± 50 , ± 100 , ± 200 bp) using the duration-plus-convexity approximation, and compare the result to full repricing — and explain the gap at the largest shocks.

4 Credit risk

4.1 Two reasons a bond can lose value

The previous chapter held one thing fixed without saying so: that the coupons **will be paid**. For a government bond in its own currency that is a safe assumption; for a corporate bond it is not. A corporate bond's yield is **higher** than a government bond's of the same maturity, and the gap — the **credit spread** — is the market's price for the risk that the borrower defaults. So a fixed-income position can lose value two ways: the **risk-free rate** can rise (interest-rate risk, last chapter), or the **spread** can widen because the borrower looks shakier (credit risk, this chapter). Understanding fixed income means holding both in view at once.

4.2 Spreads, ratings, default sensitivity

Definition 6 (credit risk and its measures) Credit risk is the risk that a borrower fails to make promised payments. It is read through three lenses:

- **credit spread** — the yield premium of a bond over a **risk-free benchmark** of the same maturity, compensating the lender for default risk; it **widens** when default looks more likely and **narrows** when it recedes;
- **ratings** — agency grades (AAA ... BBB = investment grade; BB ... D = high yield) that map to historical **default probabilities**;
- **default sensitivity** — how the bond's value responds to a change in the **probability of default** and the **recovery rate** (the fraction repaid in default).

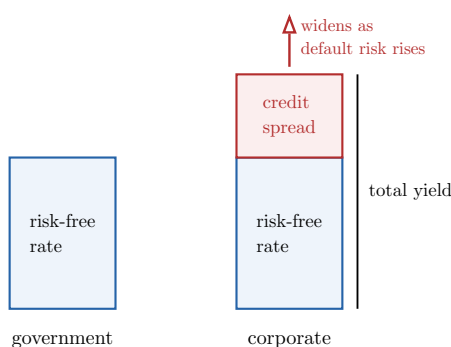


Figure 3: Yield decomposition. A corporate bond's yield is the **risk-free rate** (shared with the government bond) **plus** a **credit spread** that pays the lender for default risk. The bond loses value if the risk-free rate rises **or** if the spread widens — and in a crisis both can move against the holder at once.

4.3 Where credit risk and rate risk meet

Definition 7 (interaction of credit and interest-rate risk) A bond's yield decomposes as $y = r_{\text{free}} + s$, where s is the credit spread. Its value is therefore exposed to **both** components: a rise in r_{free} (interest-rate risk) and a widening of s (credit risk). The two are **not independent**. In a **flight to quality** — a recession or panic — investors sell risky debt and buy government bonds, so r_{free} **falls** while s **widens**; the two partly offset for a corporate bond. In a stagflationary or sovereign-stress episode they can move **together against** the holder. The net effect on a fixed-income position depends on which force dominates.

Worked example — Same rating, different exposure. Two BBB bonds yield 5%: one a 2-year note, one a 20-year note. A 100 bp **spread** widening hits both through credit risk, but the 20-year bond, with far higher **duration**, also loses far more from any **rate** move. A credit-driven sell-off that simultaneously pushes rates up devastates the long bond and barely scratches the short one — identical credit quality, opposite total risk, because duration multiplies **both** sources of yield change.

Common pitfall Do not equate **rating** with **risk**. A rating speaks to **default probability** only; it says nothing about **duration** (rate sensitivity) or **spread volatility**. A AAA 30-year bond can be far riskier in price terms than a BB 2-year bond, because its long duration dominates. In fixed income, always read **credit quality and interest-rate sensitivity** as **two separate axes** — never collapse them into one number.

Exercises

7. For a real corporate bond, decompose its yield into the risk-free benchmark and the credit spread, locate it on the rating-to-default-probability scale, and estimate its default sensitivity to a change in default probability and recovery.
8. Construct a scenario in which credit and interest-rate risk **offset** and one in which they **compound**, and quantify the net effect on a chosen bond using its duration and a spread shock.

5 Derivatives

5.1 Contracts whose value comes from somewhere else

Equities and bonds are claims you **own**. A derivative is a **contract about** something else — a bet, a promise, or an insurance policy whose payoff is written in terms of an **underlying** asset's price. Three families dominate, and each answers a different need: **futures** let you lock a price today for a trade later; **options** give you the right without the obligation, so you keep the upside and cap the downside; **swaps** let you exchange one kind of cash-flow stream for another. What unites them is a powerful pricing idea: **if a derivative could be replicated by trading the underlying, its price is pinned by the cost of that replication** — otherwise a riskless arbitrage exists.

5.2 Futures: pricing by arbitrage

Definition 8 (futures and cost-of-carry pricing) A **futures (or forward) contract** obliges the holder to buy or sell an underlying at a **fixed price** on a future date. Its fair price is set by **arbitrage**: buying the asset now and storing it must cost the same as agreeing to buy it later. With continuous compounding, no income, and risk-free rate r ,

$$F_0 = S_0 \cdot e^{rT},$$

where S_0 is the spot price and T the time to delivery. If the market futures price deviates from F_0 , a trader earns a **riskless profit** by buying the cheap leg and selling the dear one — so the deviation cannot persist.

Worked example — Closing an arbitrage. Gold spot is \$2,000, $r = 4\%$, $T = 1$ year. Fair futures: $2000 \times e^{0.04} \approx \$2,082$. Suppose the futures trades at \$2,150. Arbitrage: **borrow** \$2,000, **buy** the gold, **sell** the future at \$2,150. In a year, deliver the gold for \$2,150, repay \$2,082 of loan-plus-interest, and pocket \$68 **risk-free**. Traders doing this push the futures price down to \$2,082, where the free lunch disappears. The formula is not a forecast of the gold price — it is the **no-arbitrage** price given today's spot and rate.

5.3 Options: the right, not the obligation

Definition 9 (options: calls and puts) An **option** gives the right, **not the obligation**, to trade the underlying at a **strike price** K at (or before) expiry T , for a **premium** paid up front.

- a **call** is the right to **buy** at K : it pays $\max(S_T - K, 0)$ at expiry — valuable when the price rises above the strike;

- a **put** is the right to **sell** at K : it pays $\max(K - S_T, 0)$ — valuable when the price falls below the strike.

The buyer's loss is **capped at the premium**; the upside is large (calls) or substantial (puts). This **asymmetry** — limited loss, open-ended gain — is the defining feature of an option.

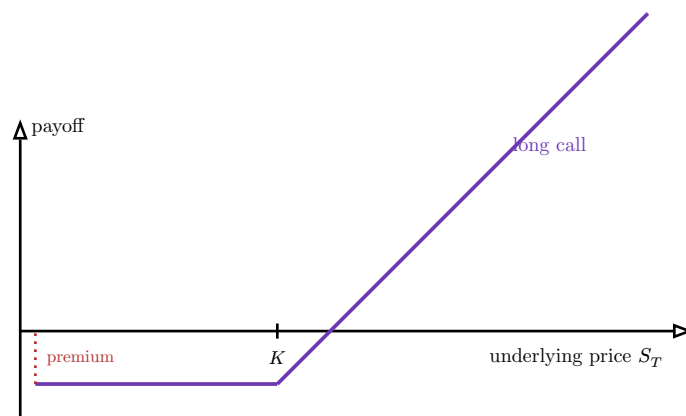


Figure 4: Profit of a long call at expiry. Below the strike K the option expires worthless and the holder loses only the **premium** paid (the flat floor): the loss is **capped**. Above K the payoff rises one-for-one with the underlying: the gain is **open-ended**. This asymmetry — the kinked payoff — is what you buy when you buy an option, and what the premium pays for.

Definition 10 (Black-Scholes intuition) The **Black-Scholes** model prices a European option from five inputs: spot S , strike K , time to expiry T , risk-free rate r , and **volatility** σ . The deep intuition is that the option can be **replicated** by a continuously-adjusted position in the underlying plus cash, so its price equals the cost of that replicating portfolio. Two consequences matter most: option value **rises with volatility** and **with time**, because both widen the range of possible outcomes while the downside stays capped at the premium — more uncertainty is **good** for the option holder. (Notably, the **expected return** of the stock does not appear: pricing is by arbitrage, not by forecasting direction.)

Definition 11 (the Greeks) The **Greeks** measure an option's sensitivities to its inputs:

- **delta** (Δ) — change in option value per unit change in the **underlying price**; it also approximates the hedge ratio and the probability of finishing in-the-money;
- **gamma** (Γ) — change in **delta** per unit change in the underlying; high gamma means the hedge must be re-adjusted often;
- **vega** (ν) — change in option value per unit change in **volatility**; long options are long vega — they gain when volatility rises.

5.4 Swaps: exchanging cash-flow streams

Definition 12 (swaps) A **swap** exchanges two streams of payments on a notional principal.

- an **interest-rate swap** exchanges **fixed** interest for **floating** (or vice versa) — used to convert a floating-rate liability into a fixed one, or to take a view on rates;

- a **currency swap** exchanges principal and interest in **one currency** for those in **another** — used to fund or hedge a foreign-currency exposure.

A swap is priced so that, at inception, the present values of the two legs are **equal** (zero net value); it then gains or loses value as rates or exchange rates move.

Common pitfall The danger of derivatives is **leverage**, and it hides in the notional. A futures or swap position controls a large notional for little or no up-front cash, so a small move in the underlying produces a large gain **or loss** relative to capital. An option **buyer's** loss is capped at the premium, but an option **seller's** (writer's) loss can be very large — selling a naked call has **unlimited** downside. Always size a derivative by the **notional exposure it creates**, not by the premium or margin it costs to enter.

Exercises

9. Explain and apply cost-of-carry futures pricing: compute the fair futures price for a real asset and identify the arbitrage trade that closes a quoted mispricing.
10. Draw the expiry payoff of a call and a put, explain the Black-Scholes inputs and why value rises with volatility and time, and interpret delta, gamma, and vega for a quoted option.
11. Explain the economic logic of an interest-rate swap and a currency swap, and describe an exposure each one would be used to transform or hedge.

6 Hedging strategy design

6.1 From understanding the instrument to deploying it

Knowing how a derivative is priced is not the same as knowing how to **use** it. A hedge is a deliberately-constructed derivative position whose payoff **moves opposite** to an existing exposure, so that gains on one offset losses on the other. Designing one on real data is a disciplined three-step craft: name the exposure precisely, choose the instrument whose payoff mirrors it, and **size** the position so the offset is neither too small (under-hedged, still exposed) nor too large (over-hedged, now a speculation in the other direction).

6.2 The three steps

Definition 13 (hedging strategy design) A **hedge** offsets the risk of an existing exposure with an opposing derivative position. Designing one requires:

- **identify the exposure** — the asset, rate, or currency whose movement creates risk; its **direction** (do you lose when it rises or falls?) and its **size**;
- **select the instrument** — the future, option, or swap whose payoff moves **against** the exposure: a forward to lock a price, an option to insure while keeping upside, a swap to convert a rate or currency;
- **calibrate the position** — set the **hedge ratio** so the offsetting payoff matches the exposure, sizing by the exposure's **delta** (for options) or **duration** (for bonds), and by notional.

Worked example — Hedging a currency receivable. A European firm will receive \$10m in three months. Its exposure: it **loses** if the dollar **weakens** against the euro. **Instrument:** sell \$10m forward (or buy a euro call / dollar put to keep upside). **Calibration:** a **forward** on the full \$10m locks the rate exactly — a one- for-one hedge ratio — removing both downside and upside. An **option** on \$10m insures the downside for a premium while keeping the gain if the dollar strengthens. The choice between them is the choice between **certainty** and **insurance with upside**.

Worked example — Hedging a bond portfolio's rate risk. A \$50m bond portfolio has modified duration 7; the manager fears rising rates. To neutralise the rate exposure, take a **short** position in interest-rate futures sized so the futures' duration-dollars offset the portfolio's: the hedge ratio is set so $D_{\text{portfolio}}^* \times \$50m \approx D_{\text{futures}}^* \times \text{not}_{\text{futures}}$. Get the ratio right and a rate rise that costs the portfolio is recovered on the short futures; get it wrong and you are still exposed — or now betting the other way.

Common pitfall Two calibration errors undo a hedge. **Over-hedging** (a hedge ratio above 1, or a notional larger than the exposure) converts a hedge into a **new speculative position** in the opposite direction — you are no longer protected, you are betting. **Basis risk** arises when the hedging instrument is not a perfect match for the exposure (e.g. hedging jet fuel with crude-oil futures): the two prices move **together but not identically**, so a residual risk remains even at the “right” size. A hedge is only as good as the match between the instrument and the exposure — name the basis risk you are accepting.

Exercises

12. For a real exposure (an FX receivable, a commodity cost, or a bond portfolio), identify the exposure's direction and size, select an appropriate hedging instrument, and calibrate the hedge ratio on the exposure data.
13. Compare a **forward** hedge and an **option** hedge for the same exposure, stating what each costs, what it removes, and what it leaves; and identify any basis risk in your chosen instrument.

7 Funds and ETFs

7.1 Buying a portfolio in one trade — but how good is it?

A fund packages a whole portfolio into a single tradeable instrument: one purchase buys diversified exposure that would otherwise take dozens of trades to assemble. The question this chapter answers is **how to judge one**. A fund is almost never evaluated in isolation — it is judged **relative to a benchmark** it claims to track or beat. From that comparison come the two numbers that matter: how **much** it deviates from the benchmark (and whether that deviation earns its fee), and how **reliably** it follows it.

7.2 Benchmark-relative performance

Definition 14 (active return and tracking error) A fund is judged against a **benchmark** index:

- **active return** (excess return) — the fund's return **minus** the benchmark's: the value the manager added or destroyed;
- **tracking error** — the **standard deviation** of the active return over time: how **closely** the fund follows the benchmark. Low tracking error means the fund hugs the index; high tracking error means it diverges, for better or worse.

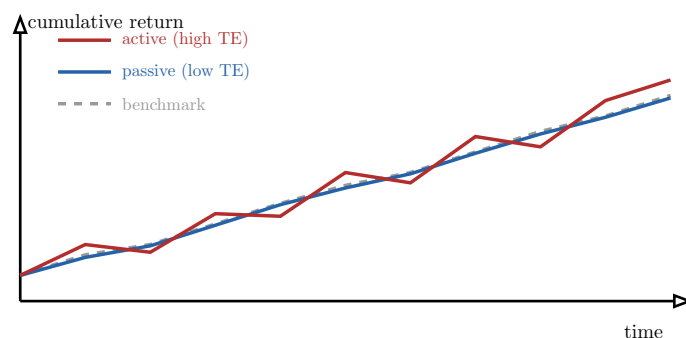


Figure 5: Tracking error visualised. The **passive** fund (blue) clings to the benchmark (grey dashed) — small, steady active return, **low** tracking error. The **active** fund (red) swings above and below it — **high** tracking error. Tracking error measures the **width of the wobble** around the benchmark, not whether the fund ends up ahead; a fund can have high tracking error and still underperform.

7.3 Active versus passive

Definition 15 (active-vs-passive efficiency and fund selection)

- **Passive** funds replicate a benchmark: near-zero active return **by design**, very low tracking error, and **low fees**.
- **Active** funds try to **beat** the benchmark: non-zero (positive or negative) active return, higher tracking error, and **higher fees**.

Efficiency asks whether an active fund's active return **justifies its fee and risk** — a manager who beats the index by 0.5% while charging 1% has destroyed value **net of fees**.

Fund-selection analytics evaluate a fund on historical data by **net** active return, tracking error, fee, and the consistency of the excess return.

Common pitfall Judge a fund **net of fees and against the right benchmark**. Two traps: comparing a fund to a benchmark it does not actually track (a small-cap fund flattered by beating a large-cap index) flatters or maligns it unfairly; and reading **gross** outperformance while ignoring the fee that the investor actually pays. A small, persistent fee compounds into a large drag over time — which is the core empirical case **for** passive investing, and exactly the question BUS63-9 takes up.

Exercises

14. For a real fund or ETF, compute its active return and tracking error against the correct benchmark on historical data, and interpret whether it behaves as active or passive.
15. Apply fund-selection analytics to choose among several funds on **net-of-fee** active return, tracking error, and fee, and justify the choice.

8 Multi-asset portfolio evaluation

8.1 The portfolio is more than the sum of its assets

Up to here each instrument was judged on its own. But an investor holds a **portfolio**, and the portfolio has properties none of its components has alone — chief among them **diversification**: combine assets that do not move together and the portfolio's risk falls **below** the average risk of its parts, a genuinely free lunch. Evaluating a multi-asset portfolio on historical data means measuring its **return and its risk together**, and reading the **correlation structure** that determines how much diversification it is actually getting.

8.2 The four evaluation metrics

Definition 16 (portfolio evaluation metrics) A multi-asset portfolio is evaluated on historical data with four numbers:

- **volatility** — the standard deviation σ_p of portfolio returns: the headline risk;
- **Sharpe ratio** — excess return per unit of risk,

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p},$$

the single best summary of risk-adjusted performance;

- **maximum drawdown** — the largest peak-to-trough fall in portfolio value over the period: the worst loss an investor living through it would have suffered;
- **cross-asset correlations** — the pairwise correlation ρ_{ij} of asset returns, which **drives the diversification benefit**.

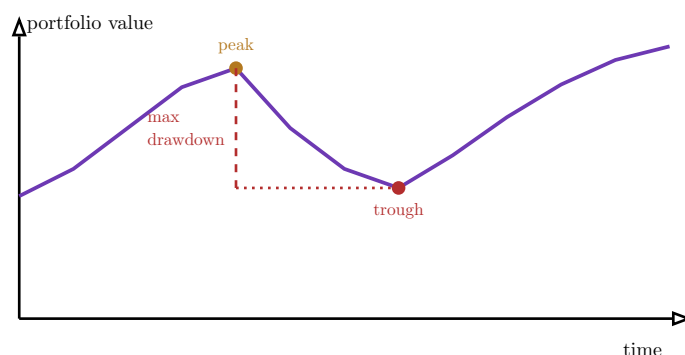


Figure 6: Maximum drawdown. It is the largest **peak-to-trough** decline (here from the peak to the later trough), not the volatility and not the final return. Two portfolios can end at the same value with the same Sharpe ratio yet have very different drawdowns — and drawdown is what tests whether an investor **holds on** or panics and sells at the bottom.

8.3 The diversification signal

Definition 17 (the diversification signal) Diversification is read from the **correlation structure**. For a two-asset portfolio,

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2.$$

Because the cross term carries ρ_{12} , **lower correlation means lower portfolio variance**. When $\rho < 1$ the portfolio's volatility is **strictly below** the weighted average of the individual

volatilities — that gap **is** the diversification benefit, and it is largest when correlations are low or negative.

Common pitfall Correlations are not constant — and they rise exactly when you need them low. The diversification measured in calm historical data routinely **evaporates in a crisis**: in a market panic, assets that normally move independently fall **together** ($\rho \rightarrow 1$), so the protection you counted on disappears at the worst moment. Always check whether a portfolio’s diversification holds in **stressed** sub-periods, not just over the full sample — a Sharpe ratio computed over a calm decade can badly overstate forward safety.

Exercises

16. Evaluate a real multi-asset portfolio on historical data: compute its volatility, Sharpe ratio, maximum drawdown, and the cross-asset correlation matrix.
17. Interpret the diversification signal from the correlation structure, and test whether it holds in a stressed sub-period as well as over the full sample.

9 Portfolio construction

9.1 From evaluating a portfolio to building one

Evaluation looks **backward** at a portfolio that exists; construction looks **forward** and **assembles** one to fit a goal. The goal is a **risk profile** — a stated tolerance for volatility and loss — and the construction must align the portfolio’s risk to it through deliberate choices of **which** assets, in **what** weights. The crowning tool is **mean-variance optimisation**, which turns the diversification mathematics of the last chapter into an explicit recipe for the best portfolio at each level of risk — and whose **practical limits** are as important to learn as its logic.

9.2 The construction decisions and the efficient frontier

Definition 18 (portfolio construction) Portfolio construction assembles a portfolio aligned with a defined **risk profile** through: **selection** (which assets to include), **sizing** (the weights), **diversification** (spreading risk across low-correlation assets), and **risk-profile alignment** (matching the resulting volatility and drawdown to the investor’s tolerance).

Definition 19 (mean-variance optimisation) Mean-variance optimisation (Markowitz) chooses weights to **minimise variance for a target expected return** — equivalently, to maximise return for a target variance. Sweeping the target traces the **efficient frontier**: the set of portfolios offering the **highest expected return for each level of risk**. Its inputs are the assets’ **expected returns**, **volatilities**, and **correlations**. The point where a line from the risk-free rate is **tangent** to the frontier is the **maximum-Sharpe** portfolio.

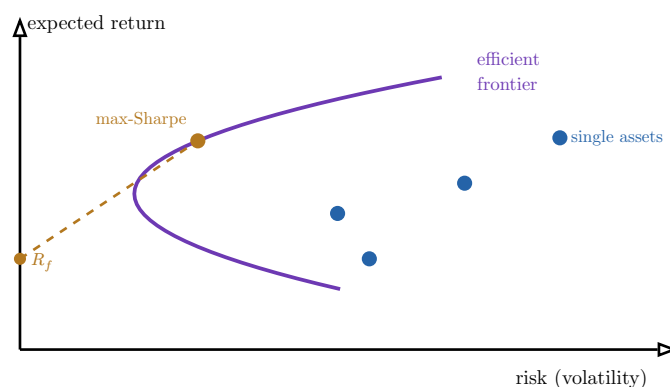


Figure 7: The efficient frontier. Every individual asset (blue) sits **inside** the frontier (purple) — dominated by some diversified mix with the same risk and higher return, or the same return and lower risk. Optimisation finds the frontier; the **risk profile** picks the point on it. The dashed line from the risk-free rate R_f touches the frontier at the **maximum-Sharpe** portfolio.

Worked example — Aligning the portfolio to the risk profile. A cautious retiree and an aggressive young saver face the **same** efficient frontier but choose **different** points on it. The retiree picks a low-volatility point — more bonds, modest equity — accepting lower expected return for a shallower worst-case drawdown. The saver picks a high-volatility point — mostly equities, perhaps leverage toward the max-Sharpe portfolio — accepting deep drawdowns for higher expected return. Construction is not finding **the** best portfolio; it is finding the best portfolio **for a stated risk profile**.

Common pitfall Mean-variance optimisation is dangerously sensitive to its inputs — state this whenever you use it. The expected returns and correlations are **estimated with error** from historical data, and the optimiser treats them as exact. The result: tiny changes in the inputs produce **large, unstable swings** in the weights, and the “optimal” portfolio often **concentrates** in a few assets — the opposite of diversification. It also chases estimation noise, loading up on whatever asset happened to look best in the sample. In practice the raw output is **constrained** (caps on weights), the inputs are **shrunk** toward sensible priors, and the optimiser is treated as a **guide, not a prescription**. An optimised portfolio reported without these caveats is a misuse of the tool.

Exercises

18. Construct and justify a multi-asset portfolio aligned with a defined risk profile, selecting and sizing assets and applying mean-variance optimisation to locate the frontier and choose a point on it.
19. Demonstrate the instability of mean-variance optimisation: perturb the expected-return inputs slightly, show how the optimal weights shift, and state the practical safeguards (constraints, shrinkage) you would apply.

10 Communicating an investment recommendation

10.1 The analysis is worthless until someone acts on it

Every technique in this book produces numbers. But the people who **allocate capital** — a client, a committee, a non-technical decision-maker — do not act on a duration or a Sharpe

ratio; they act on a **recommendation they understand and trust**. The final competence of the course is therefore **communication**: turning the quantitative work into a clear, honest, visual narrative that states **what** to do, **how much**, and **why**, in language the audience can act on. A correct analysis that no one understands changes nothing.

10.2 Visualisation and narrative

Definition 20 (the investment recommendation) An investment recommendation presents a constructed portfolio to a **non-technical audience** through:

- **data visualisation** (Python / matplotlib) — charts of **allocation** (what is held), **historical performance** (how the mix behaved), and **risk** (volatility, drawdown), each chosen to make one point clearly rather than to display every number;
- **narrative structure** — a written rationale covering **return expectations**, **risk management**, and the **data sources used**.

It states the portfolio **selected**, how it is **sized**, and **why**, translating the analysis into a decision the audience can act on.

Worked example — Two ways to present the same portfolio. Version A: “The mean-variance optimiser produced weights of 0.42, 0.31, 0.18, 0.09 with an ex-ante Sharpe of 0.84 and annualised σ of 11.3%.” Version B: “I recommend roughly 40% global equities, 30% bonds, 20% gold, and 10% cash. Over the past decade this mix returned about 7% a year, with its worst loss around 18% — milder than the 35% an all-equity portfolio suffered. It fits your goal of steady growth without sleepless nights.” Same portfolio. Version A informs no decision; Version B **is** the recommendation — quantities, behaviour, and rationale in the audience’s language.

Common pitfall The twin failures of investment communication are **opposite**. **Over-technical**: burying the audience in Greeks, frontiers, and basis points until they cannot act — sophistication as a substitute for clarity. **Over-confident**: presenting a single number or back-test as a promise, hiding the assumptions, the estimation error, and the downside. Honest communication does both jobs at once — **simple enough to act on, candid enough to trust** — which means stating the return **expectation** as a range, the risk in terms the audience feels (the drawdown they would have lived through), and the **data sources and their limits** plainly.

Exercises

20. Produce a complete data-driven investment recommendation for a non-technical audience: select and size a portfolio, build matplotlib visualisations of allocation, performance, and risk, and write a rationale covering return expectations, risk management, and data sources.

21. Take a technically-correct but jargon-heavy recommendation and rewrite it for a non-technical client, preserving every material fact — including the downside and the data limits — while removing the jargon.

Financial instruments, in one paragraph. You arrived able to price a **firm**; you leave able to price the **instruments** through which a view on a firm — or a market — is actually bought and sold. Sort every instrument into its **family** — equity (residual, high risk), debt (contractual, senior), derivative (value derived from an underlying, used to hedge or lever), fund (pooled, active or passive) — by its place in the capital structure and on the risk-return map. Compress an equity's fundamentals into a **quantified score** that ranks a universe. Price a bond as the present value of fixed cash at its **YTM**, and measure its rate risk with **duration** and the **convexity** correction that duration alone gets wrong for large shocks. Read fixed income on **two axes at once** — interest-rate risk and the **credit spread** — knowing they can offset or compound. Price **futures** by no-arbitrage cost-of-carry, understand **options** through their capped-loss / open-upside asymmetry and the Black-Scholes intuition (value rises with volatility and time), read the **Greeks**, and use **swaps** to transform a rate or currency stream. **Hedge** by naming the exposure, selecting the offsetting instrument, and **calibrating the hedge ratio** — wary of over-hedging and basis risk. Judge **funds** against the right benchmark on **net** active return and **tracking error**, and weigh active against passive. Evaluate a **portfolio** on volatility, **Sharpe**, **maximum drawdown**, and the **correlation structure** that drives diversification — which weakens exactly in a crisis. **Construct** a portfolio to a **risk profile** with **mean-variance optimisation**, always stating its instability and treating it as a guide, not a prescription. Then **communicate** the result to a non-technical audience — visual, quantified, and honest about the downside and the data. One reflex throughout: **what are its cash flows, what moves its price, and what does it do to the portfolio I already hold?**